

Exp No: 27 BINARY SEMAPHORE IN SYSTEMC

DATE: 25/08/2026

```
kaviya@KV: ~
GNU nano 8.7.1
#include <systemc.h>

SC_MODULE(binary_semaphore) {

    int sem;
    sc_event ev;

    void task1() {
        while(true) {
            wait(1, SC_SEC);

            if(sem == 1) {

                sem = 0;
                cout << "Task1 Entered Critical Section at " << sc_time_stamp() << endl;
                wait(2, SC_SEC);
                sem = 1;
                ev.notify();
            }
            else {
                wait(ev);
            }
        }
    }

    void task2() {
        while(true) {
            wait(2, SC_SEC);

            if(sem == 1) {
                sem = 0;
                cout << "Task2 Entered Critical Section at " << sc_time_stamp() << endl;
                wait(1, SC_SEC);
                sem = 1;
                ev.notify();
            }
            else {
```

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else {
wait(ev);
}
}
}

void task2() {
while(true) {
wait(2, SC_SEC);

if(sem == 1) {
sem = 0;
cout << "Task2 Entered Critical Section at " << sc_time_stamp() << endl;
wait(1, SC_SEC);
sem = 1;
ev.notify();
}
else {
wait(ev);
}
}
}

SC_CTOR(binary_semaphore) {
sem = 1;
SC_THREAD(task1);
SC_THREAD(task2);
}

};

int sc_main(int argc, char* argv[]) {
binary_semaphore obj("Binary_Semaphore");
sc_start(10, SC_SEC);
return 0;
}
|
```

```
kaviya@KV: ~
kaviya@KV:~$ nano exp27.cpp
kaviya@KV:~$ g++ -o exp27 exp27.cpp -lsystemc
kaviya@KV:~$ ./exp27

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
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ALL RIGHTS RESERVED
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
kaviya@KV:~$ |
```